Monitoring Report

CARBON OFFSET UNIT (CoU) PROJECT

Title: 5.5 MW Bundled Wind Power Project in Gujarat, India

Version 1.2

Date: 31/07/2025

First CoU Issuance Period: 5 years, 8 months, 14 days

Monitoring Period: 17/04/2019 to 31/12/2024

Monitoring Report (MR)
CARBON OFFSET UNIT (CoU) PROJECT

Monitoring Report	
Title of the project activity	5.5 MW Bundled Wind Power Project in Gujarat, India
UCR Project Registration Number	451
Version	1.2
Completion date of the MR	31/07/2025
Monitoring period number and duration of this monitoring period	Monitoring Period Number: 01 Duration of this monitoring Period: 5 years, 8 months, 14 days (first and last days included) 17/04/2019 to 31/12/2024
Project participants	**Representator:** Advait Greenergy Private Limited **Developer:** M/s. Kutch Chemical Industries Ltd. M/s. Panoli Intermediates (India) Pvt. Ltd.
Host Party	India
Applied methodologies and standardized baselines	Applied Methodologies: UNFCCC Approved Small Scale Methodology "AMS-I.D., Grid connected renewable electricity generation", Version – 18.0 Standardized Baselines: N/A
Sectoral scopes	01 Energy industries (Renewable/NonRenewable Sources)
Actual amount of GHG emission reductions for this monitoring period	2019: 5,811 CoUs (5,811 tCO_{2eq})
	2020: 6,008 CoUs (6,008 tCO_{2eq})
	2021: 7,059 CoUs (7,059 tCO_{2eq})
	2022: 5,546 CoUs (5,546 tCO_{2eq})
	2023: 3,872 CoUs (3,872 tCO_{2eq})
	2024: 4,412 CoUs (4,412 tCO_{2eq})
Total:	**32,708** CoUs (**32,708** tCO_{2eq})

SECTION A. Description of project activity

A.1. Purpose and general description of project activity >>

a) Purpose of the project activity and the measures taken for GHG emission reductions >>

The project activity is a renewable power generation initiative involving the installation and operation of 4 Wind Turbine Generators (WTGs) having capacity 1x1.5 MW, 1x1.5 MW, and 2x1.25 MW each manufactured and supplied by Suzlon Energy with aggregated installed capacity of 5.5 MW in Kutch district, Gujarat, India. This project has been developed and promoted by M/s. Panoli Intermediates (India) Pvt. Ltd. and M/s. Kutch Chemical Industries Limited to generate clean energy and reduce greenhouse gas (GHG) emissions.

The WTGs under the project activity were commissioned as per the below table:

Project Developer	Capacity of WTG	Commissioning Date	Location
M/s. Panoli Intermediates (India) Pvt. Ltd.	1x1.5 MW	14-Feb-08	Village: Nani Sindholi Taluka: Abdasa District: Kutch State: Gujarat Country: India
M/s. Kutch Chemical Industries Limited	1x1.5 MW	14-Feb-08	Village: Raparghad Taluka: Abdasa District: Kutch State: Gujarat Country: India
M/s. Kutch Chemical Industries Limited	2x1.25 MW	29-Sep-08	Village: Bada & Panchetiya Taluka: Mandvi District: Kutch State: Gujarat Country: India

The generated power is wheeled through the Indian grid (formerly known as the NEWNE grid)[1] to the manufacturing facility of the project proponent (PP) in Gujarat for captive consumption, in accordance with the wheeling agreement signed between Gujarat Energy Transmission Corporation Limited (GETCO) and the PP. The actual net electricity generated and supplied to the grid during this period was 37,303.691 MWh; with annual electricity generation of 540.63 MWh.

By displacing grid electricity that would have otherwise been generated by fossil fuel-based power plants, the project achieved an actual greenhouse gas (GHG) emission reduction of 32,708 tCO_2 during the monitoring period. The use of wind energy, a zero-emission source, has directly contributed to reducing reliance on fossil fuel-based power generation in the grid.

[1] National Grid

b) Brief description of the installed technology and equipment>>

The project activity involves the installation and operation of four Wind Turbine Generators (WTGs) manufactured and supplied by Suzlon Energy, with a total installed capacity of 5.5 MW in Gujarat, India. Each WTG is connected to a Central Monitoring Station (CMS) via a high-speed WLAN modem or fiber optic cable, enabling real-time status monitoring with an intuitive Graphical User Interface (GUI). This system allows remote tracking and management of turbine performance, ensuring efficient operation.

The project also incorporates a Supervisory Control & Data Acquisition (SCADA) system, which provides a graphical representation of operational data, long-term data storage, and historical analysis. It facilitates access to daily generation reports and power curve monitoring while enabling both real-time and offline troubleshooting with advanced analytical tools. Additionally, the WTGs are equipped with a safety system and instrumentation to track individual turbine functions, ensuring operational efficiency and reliability. As per manufacturer specifications, the expected operational lifetime of each WTG is 20 years.

c) Relevant dates for the project activity (e.g. construction, commissioning, continued operation periods, etc.)>>

The duration of the crediting period corresponding to the monitoring period covered in this monitoring report.

UCR Project ID or Date of Authorization : 451
Start Date of Crediting Period : 17/04/2019
Project Commissioned : 14/02/2008

d) Total GHG emission reductions achieved or net anthropogenic GHG removals by sinks achieved in this monitoring period>>

The total GHG emission reductions achieved in this monitoring period is as follows:

ERs Generated for the Monitoring Period	
Start date of this Monitoring Period	17/04/2019
Carbon credits claimed up to	31/12/2024
Total ERs generated (tCO_{2eq})	32,708 tCO_{2eq}
Leakage	0 tCO_{2eq}

e) Baseline Scenario>>

In the absence of the project activity, the equivalent amount of electricity would have been generated by power plants connected to the Indian grid, which are predominantly fossil fuel-based. As a result, the baseline scenario for the project activity is grid-based electricity generation, which represents the pre-project condition. Since the project activity utilizes wind energy for power generation, it operates as a zero-emission system and does not contribute to greenhouse gas emissions. By replacing fossil fuel-based electricity with renewable wind power, the project effectively reduces emissions that would have otherwise been released into the atmosphere.

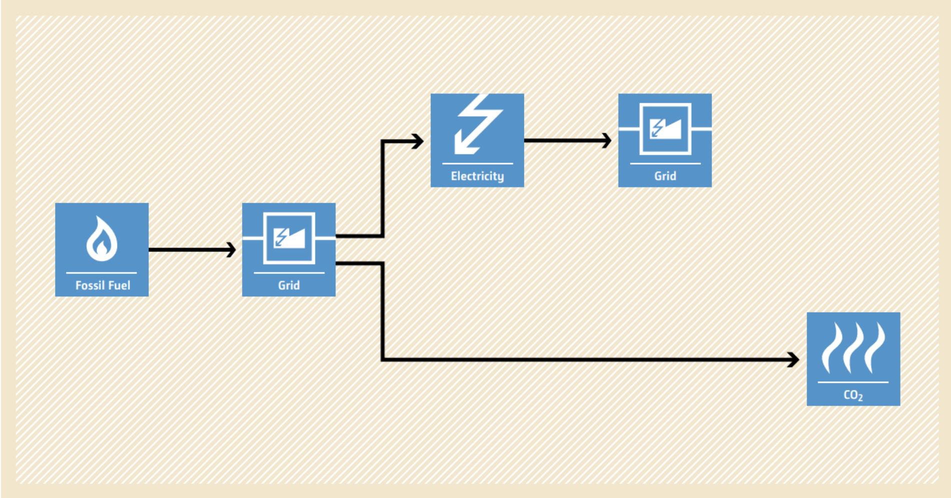

Figure 1: Baseline Scenario

A.2. Location of project activity>>

Project Developer	M/s. Panoli Intermediates (India) Pvt Ltd.	M/s. Kutch Chemical Industries Limited	M/s. Kutch Chemical Industries Limited	
Capacity	1x1.5 MW	1x1.5 MW	1x1.25 MW	1x1.25 MW
WTG Location No.	M-523	M-438	B-01	B-480
District	Kutch	Kutch	Kutch	Kutch
Village	Nani Sindholi	Raparghad	Bada	Panchetiya
Taluka	Abdasa	Abdasa	Mandvi	Mandvi
State	Gujarat	Gujarat	Gujarat	Gujarat
Country	India	India	India	India
Pin Code	370655	370655	370475	370475
Latitude	23.10626° N	23.0853° N	22.88445° N	22.86983° N
Longitude	68.80805° E	68.79203° E	69.15537° E	69.18707° E
Survey No.	63	141	526p	174/2

The representative location map is included below:

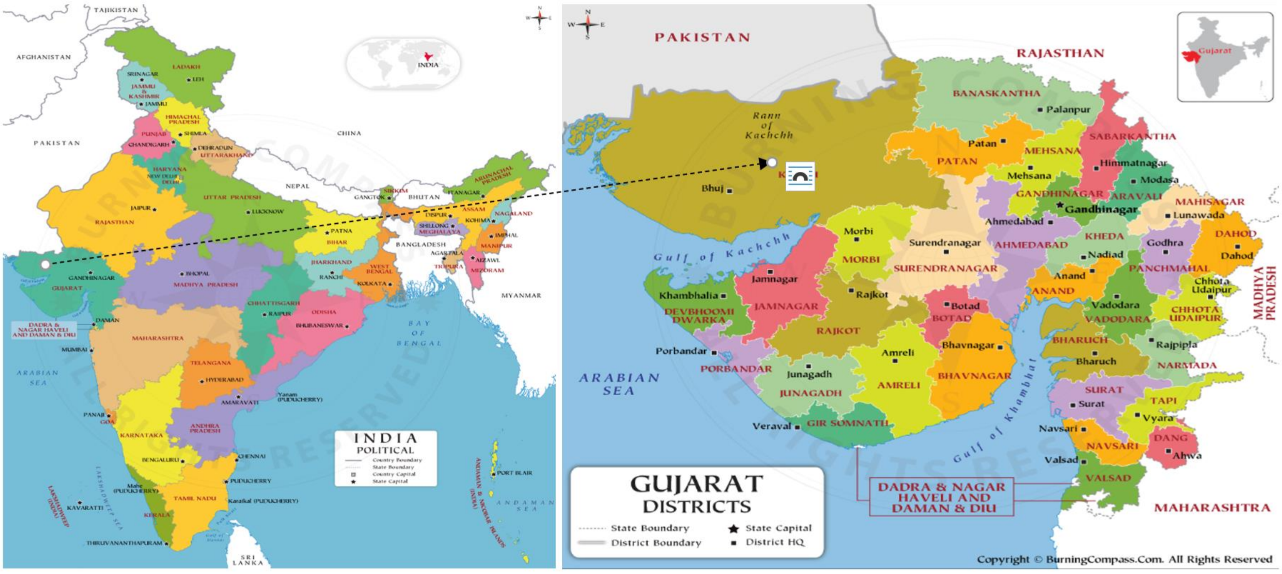

Figure 2: Project Location

(Courtesy: google images, www.burningcompass.com)

A.3. Parties and project participants >>

Party (Host)	Participants
Government of India	**Representator:** Advait Greenergy Private Limited (Representator) **Project Proponent (Developer):** M/s. Kutch Chemical Industries Ltd. M/s. Panoli Intermediates (India) Pvt. Ltd.

A.4. References to methodologies and standardized baselines >>

SECTORAL SCOPE: 01, Energy industries (Renewable/Non-renewable sources)

TYPE: I–Renewable Energy Projects

CATEGORY: AMS-I.D., Grid connected renewable electricity generation, Version 18.0[2]

A.5. Crediting period of project activity >>

This monitoring report covers 1st monitoring period.
Length of the crediting period corresponding to this monitoring period: 5 years, 8 months, 14 days (17/04/2019 to 31/12/2024)

A.6. Contact information of responsible persons/entities >>

Representator	
Name	Advait Greenergy Private Limited
Contact Person	Ms. Avantika Gupta

[2] 2P7FS6ZQAR84LG3NMKYUH50WI9ODBC (unfccc.int)

Mobile	+91 9079765066
E-mail	avantika.gupta@advaitgroup.co.in
Address	Advait Energy Transitions Limited, 1st Floor, KIFS Corporate House, Iskcon Ambli Road, Beside Hotel Planet Landmark, Near Ashok Vatika, Ambli, Ahmedabad – 380058
Project Developer	
Name	M/s. Panoli Intermediates (India) Pvt Ltd.
Contact Person	Mr. Anupam Chaturvedi
Mobile	+91 9737586359
E-mail	anupam@kcil.co.in
Address	"Saraniwas" 20-21, Hari Nagar Co-Op - Society, Gotri Road, Vadodara - 390 007, Gujarat (INDIA)
Project Proponent (Developer)	
Name	M/s. Kutch Chemical Industries Ltd.
Contact Person	Mr. Anupam Chaturvedi
Mobile	+91 9737586359
E-mail	anupam@kcil.co.in
Address	"Saranivas" 20-21, Hari Nagar Co-Op - Society, Gotri Road, Vadodara - 390 007, Gujarat (INDIA)

SECTION B. Implementation of project activity

B.1. Description of implemented registered project activity >>

a) Provide information on the implementation status of the project activity during this monitoring period in accordance with UCR PCN>>

The project activity is installation and operation of 4 Wind Turbine Generators (WTGs) manufactured and supplied by Suzlon Energy with an aggregate installed capacity of 5.5 MW in the state of Gujarat state of India.

Project Developer	Capacity of WTG	Commissioning Date	Location	Status
M/s. Panoli Intermediates (India) Pvt. Ltd	1x1.5 MW	14-Feb-08	Village: Nani Sindholi Taluka: Abdasa District: Kutch State: Gujarat Country: India	Operational
M/s. Kutch Chemical Industries Ltd.	1x1.5 MW	14-Feb-08	Village: Raparghad Taluka: Abdasa District: Kutch State: Gujarat Country: India	Operational
M/s. Kutch Chemical Industries Ltd.	2x1.25 MW	29-Sep-08	Village: Bada & Panchetiya Taluka: Mandvi District: Kutch State: Gujarat Country: India	Operational

b) For the description of the installed technology(ies), technical process and equipment, include diagrams, where appropriate>>

The project activity has been successfully implemented and is fully operational during the monitoring period. It consists of four Wind Turbine Generators (WTGs) manufactured and supplied by Suzlon Energy, with a total installed capacity of 5.5 MW in Gujarat, India.

Technical details for WTG Machine at the sites are as follows:

Project Developer	M/s. Panoli Intermediates (India) Pvt. Ltd.	M/s. Kutch Chemical Industries Limited	M/s. Kutch Chemical Industries Limited	M/s. Kutch Chemical Industries Limited
Capacity	1x1.5 MW	1x1.5 MW	1x1.25 MW	1x1.25 MW
WTG Location No.	M-523	M-438	B-01	B-480
WTG ID Number	SEL/1500/07-08/0881	SEL/1500/07-08/0880	SEL/1250/08-09/1306	SEL/1250/08-09/1307
WTG Make	Suzlon	Suzlon	Suzlon	Suzlon
Meter No.	GJU04440	XD501471	GJB01111	GJB01377

Type/Specification	Details
General Data	
Model	S82 – 1.5 MW
Power Rating	1.5 MW
Rotor Diameter	82 m
Wind Class	IEC Class IIIa
Off Shore Model	No
Swept Area	5,281 m²
No. of Blades	3
Rotor	
Minimum Rotational Speed	~15.6 rpm
Maximum Rotational Speed	~16.3 rpm
Cut-in Wind Speed	4 m/s
Rated Wind Speed	12 m/s
Cut-off Wind Speed	20 m/s
Survival Wind Speed	52.5 m/s
Blade	
Material	Epoxy bonded fiberglass
Gearbox	
Type	1 planetary stage + 2 parallel stages
Stage	3
Gear Ratio	1:95
Generator	
Generator Type	Asynchronous
Speed at Rated Power	1,511
Number	1
Voltage	690 V
Frequency	50 Hz
Insulation Class	Class H
Tower	
Tower Type	Tubular Steel
Hub Height	76.1 m
Type	Steel Tube

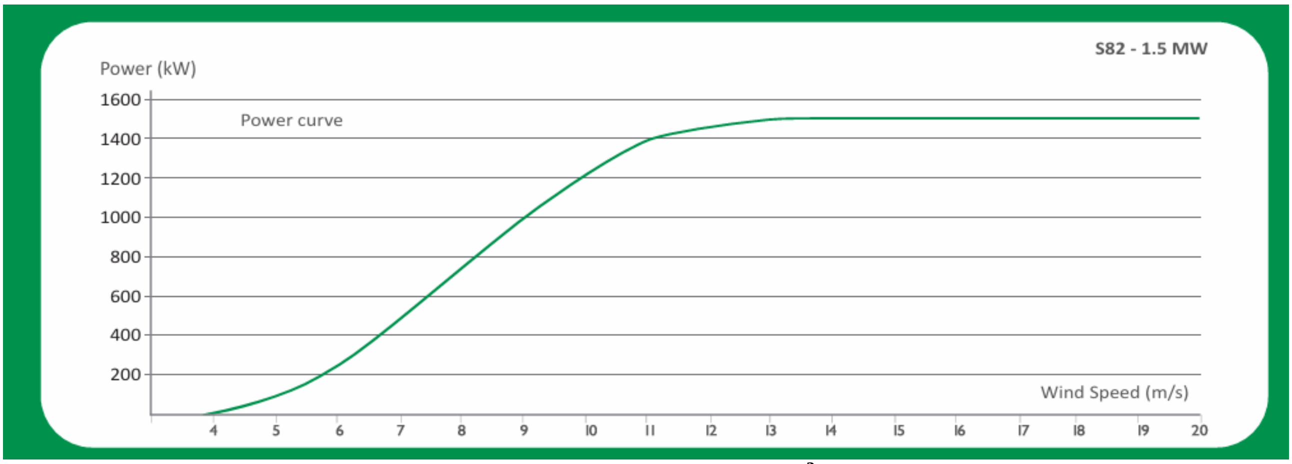

Figure 3: WTG Power Curve[3]

[3] S82 WTG

Project Developer	M/s. Kutch Chemical Industries Limited
Capacity	1x1.25 MW
WTG Location No.	B-01
WTG ID Number	SEL/1250/08-09/1306
WTG Make	Suzlon
Meter Number	GJB01111

Project Developer	M/s. Kutch Chemical Industries Limited
Capacity	1x1.25 MW
WTG Location No.	B-480
WTG ID Number	SEL/1250/08-09/1307
WTG Make	Suzlon
Meter Number	GJB01377

Type/Specification	**Details**
General Data	
Model	S66 – 1.25 MW
Power Rating	1.25 MW
Rotor Diameter	66 m
Wind Class	IEC Class III
Off Shore Model	No
Swept Area	3,421 m²
No. of Blades	3
Power Control	Pitch regulated
Rotor	
Minimum Rotational Speed	~13.5 rpm
Maximum Rotational Speed	~20.3 rpm
Cut-in Wind Speed	4 m/s
Rated Wind Speed	12 m/s
Cut-off Wind Speed	20 m/s
Survival Wind Speed	52.5 m/s
Blade	
Material	Epoxy bonded fiberglass
Gearbox	
Type	1 planetary stage + 2 parallel stages
Stage	3
Gear Ratio	1:74.9
Generator	
Generator Type	Asynchronous
Speed at Rated Power	1,006–1,506 rpm (generator side)
Number	1
Voltage	690 V
Frequency	50 Hz
Cooling System	Air cooled
Insulation Class	Class H
Tower	
Tower Type	Tubular steel

Hub Height	74.5 m
Type	Steel Tube

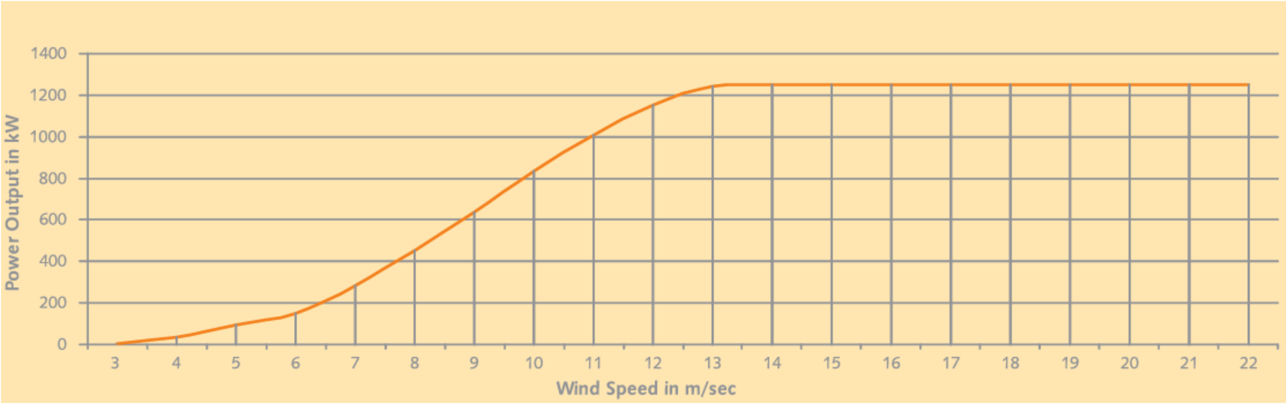

Figure 4: WTG Power Curve[4]

The WTGs have continuously generated and supplied renewable electricity to the manufacturing facility of the project proponent (PP) in Gujarat for captive consumption, as per the wheeling agreement with Gujarat Energy Transmission Corporation Limited (GETCO).

To ensure efficient operation and real-time monitoring, all WTGs are connected to a Central Monitoring Station (CMS) via a high-speed WLAN modem or fiber optic cable. This connectivity enables real-time status updates of the turbines through a Graphical User Interface (GUI), allowing seamless monitoring and management of their performance.

Additionally, a Supervisory Control & Data Acquisition (SCADA) system has been integrated, providing a graphical representation of operational data, long-term data storage, and historical analysis. The system enables daily generation reporting, power curve monitoring, and facilitates both real-time and offline troubleshooting using advanced analytical tools.

The WTGs are also equipped with a safety system and instrumentation that track individual turbine functions, ensuring operational reliability. As per manufacturer specifications, the expected operational lifetime of each WTG is 20 years.

B.2 Do no harm or Impact test of the project activity>>

There was no harm identified from the implementation and operation of the project activity; therefore, no mitigation measures are applicable.

Rational: As per the Central Pollution Control Board (MoEFCC, Govt. of India) classification of industrial sectors under Red, Orange, Green, and White Categories (07/03/2016)[5], wind power projects fall under the "**White Category**." Projects in this category are considered environmentally benign and do not require Environmental Clearance or a 'Consent to Operate' from the Pollution Control Board (PCB), as they do not cause any negative environmental impact.

Furthermore, under Indian regulations, Environmental and Social Impact Assessments (ESIA) are not required for wind energy projects.

[4] S66 WTG
[5] Environment Ministry

SDG	Relevant SDG Target	Project Contributions
SDG7: Affordable and Clean Energy	Target 7.2: By 2030, increase substantially the share of renewable energy in the global energy mix.	The project contributes to Sustainable Development Goal 7 (Affordable and Clean Energy) by generating renewable electricity using wind turbine generators. This clean energy generation displaces electricity that would otherwise be sourced from fossil fuel-based grid supply, thereby enhancing the share of renewable energy in the regional energy mix. The project will generate a total 37,303.69 MWh of electricity, directly supporting the transition to a more sustainable, reliable, and environmentally friendly energy infrastructure.
SDG13: Climate Action	Target 13.2: Integrate climate change measures into national policies, strategies and planning.	The project also contributes to Sustainable Development Goal 13 (Climate Action) by achieving measurable and verifiable reductions in greenhouse gas (GHG) emissions. By replacing carbon-intensive grid electricity with wind-generated renewable power for captive consumption at the manufacturing unit, the project resulted in total emission reduction of approximately 32,708 tonnes of CO_2 equivalent (tCO_2e). This aligns with national and international climate goals by promoting low-carbon development pathways and strengthening climate change mitigation efforts.

Beyond the absence of negative impacts, the project activity delivers social, environmental, economic, and technological benefits that contribute to sustainable development.

<u>Social well-being:</u>
The project would help in generating direct and indirect employment benefits accruing out of ancillary units for manufacturing towers for erection of the Wind Turbine Generator (WTG) and for maintenance during operation of the project activity. It will lead to development of infrastructure around the project area in terms of improved road network etc. and will also directly contribute to the development of renewable infrastructure in the region.

<u>Economic well-being:</u>
The project is a clean technology investment decided based on carbon revenue support, which signifies flows of clean energy investments into the host country. The project activity requires temporary and permanent, skilled and semi-skilled manpower at the project location; this will create

additional employment opportunities in the region. The generated electricity will be utilized for captive consumption, thereby reducing the demand from the grid. In addition, improvement in infrastructure will provide new opportunities for industries and economic activities to be setup in the area. Apart from getting better employment opportunities, the local people will get better prices for their land, thereby resulting in overall economic development.

Technological well-being:
The project activity employs state of art technology WTGs which has high power generation potential with optimized utilization of land. The successful operation of project activity would lead to promotion of this technology and would further push R&D efforts by technology providers to develop more efficient and better machinery in future. Hence, the project leads to technological well-being.

Environmental well-being:
The project activity will generate power using zero emissions wind-based power generation facility which helps to reduce GHG emissions and specific pollutants like SOx, NOx, and SPM associated with the conventional thermal power generation facilities. The project utilizes wind energy for generating electricity which is a clean source of energy. The project activity will not generate any air pollution, water pollution or solid waste to the environment which otherwise would have been generated through fossil fuels. Thus, the project causes no negative impact on the surrounding environment contributing to environmental well-being.

With regards to ESG credentials:
At present specific ESG credentials have not been evaluated, however, the project essentially contributes to various indicators which can be considered under ESG credentials. Some of the examples are as follows:

<u>Under Environment:</u>
Environmental criteria may include a company's energy use, waste, pollution, natural resource conservation, and treatment of animals etc. For PP, energy use pattern is now based on renewable energy due to the project and it also contributes to GHG emission reduction and conservation of depleting energy sources associated with the project baseline. Also, the criteria can be further evaluated on the basis of any environmental risks which the company might face and how those risks are being managed by the company. Here, as the power generation will be based on wind power, the risk of environmental concerns associated with non-renewable power generation and risk related to increasing cost of power etc. are now mitigated. Hence, project contributes to ESG credentials.

<u>Under Social:</u>
Social criteria reflect on the company's business relationships, qualitative employment, working conditions with regard to its employees' health and safety, interests of other stakeholders' etc. With respect to this project, the PP has robust policies in place to ensure equitable employment, health & safety measures, local jobs creation etc. Also, the organizational CSR activities directly support local stakeholders to ensure social sustainability. Thus, the project contributes to ESG credentials.

<u>Under Governance:</u>
Governance criteria relate to overall operational practices and accounting procedure of the organization. With respect to this project, the Project Proponent practices a good governance practice with transparency, accountability and adherence to local and national rules & regulations etc. This can be further referred from the company's annual report. Also, the project activity is a wind power project owned and managed by the PP for which all required NOCs and approvals are received. The electricity generated from the project can be accurately monitored, recorded and further verified under the existing management practice of the company. Thus, the project and the proponent ensure good credentials under ESG.

B.3. Baseline Emissions>>

Project activity installs the wind power project at a barren land. Project activity is the installations of green field energy production with the installation of 4 WTGs totalling 5.5 MW project capacity.

In the absence of the project activity the equivalent amount of electricity would have been generated from the connected/ new power plants in the Indian grid, which are/ will be predominantly based on fossil fuels[6], hence baseline scenario of the project activity is the grid-based electricity system, which is also the pre-project scenario. Since the project activity involves power generation from wind, it does not emit any emissions in the atmosphere.

Project activity will harness wind as a source of energy production which is environmentally safe and sound technology. There is no GHG emission through project activity. The WTGs confirms to the relevant code of safety and standards mandatory for setting up wind projects. The standard includes Wind Turbine Safety and Design, Noise level and Mechanical Load. Therefore, the technology implemented can be depicted as environmentally safe and sound one.

Schematic diagram showing the baseline scenario:

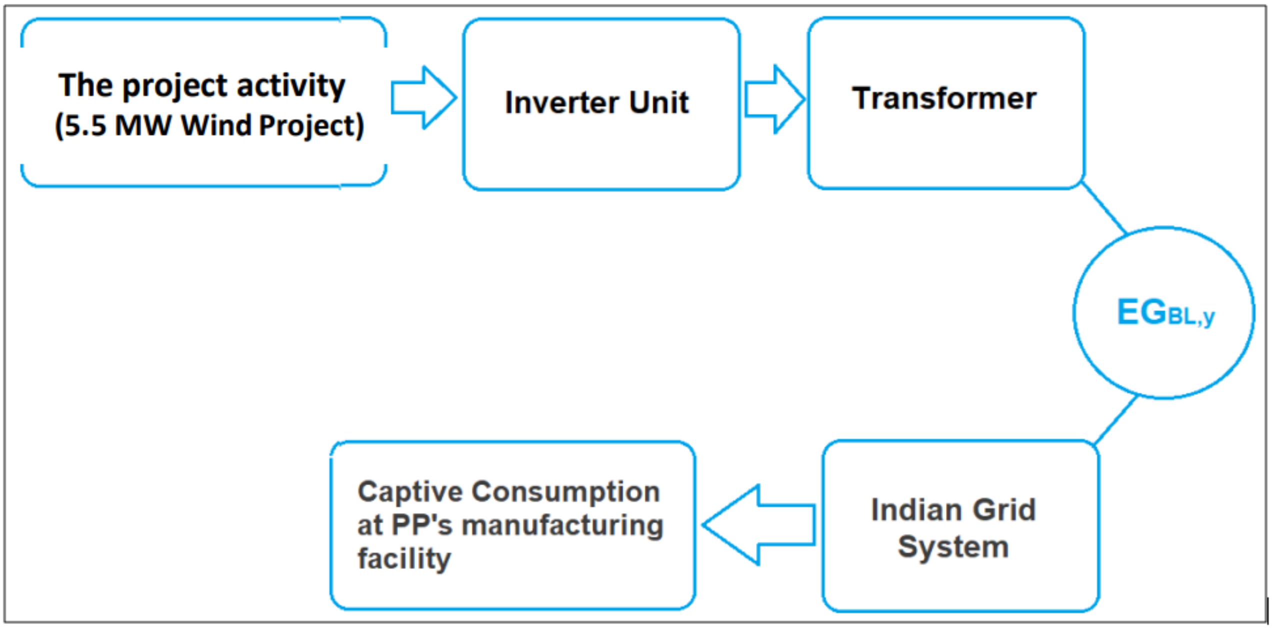

Figure 5: Project Scenario

[6] http://www.cea.nic.in/installed_capacity.html

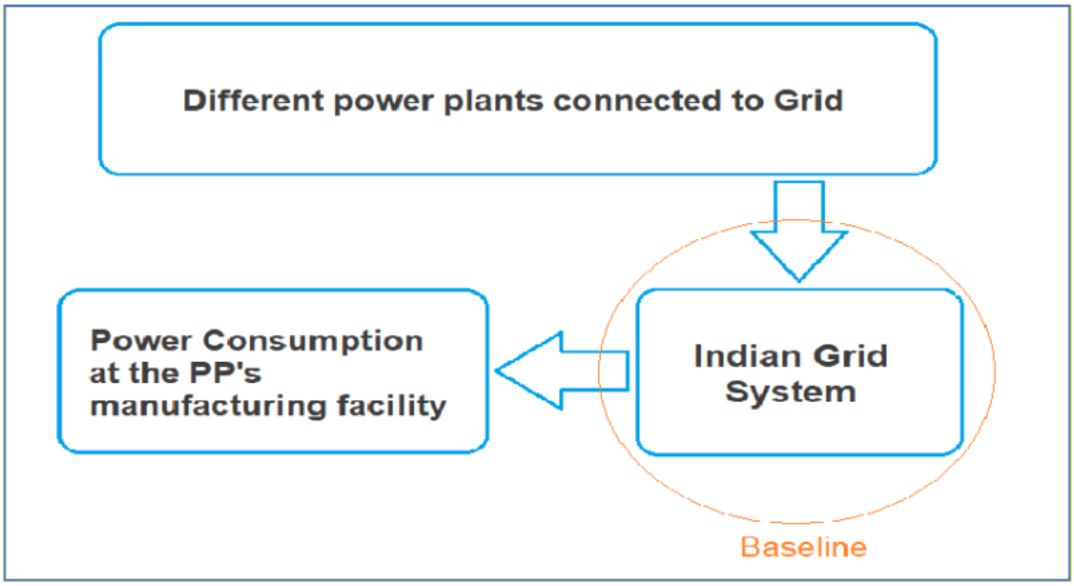

Figure 6: Baseline Location

Thus, this project activity was a voluntary investment which replaced equivalent amount of electricity from the Indian grid. The project proponent was not bound to incur this investment as it was not mandatory by national and sectoral policies. Thus, the continued operation of the project activity would continue to replace fossil fuel-based power plants and fight against the impacts of climate change. The Project Proponent hopes that carbon revenues from 17/04/2019 to 31/12/2024 accumulated as a result of carbon credits generated will help repay the loans and help in the continued maintenance of this project activity.

B.4. Debundling>>

The project activity 5.5 MW Bundled Wind Power Project in Gujarat, India by M/s. Panoli Intermediates (India) Pvt. Ltd. and M/s. Kutch Chemical Industries Limited is not a debundled component of a larger project activity.

SECTION C. Application of methodologies and standardized baselines

C.1. References to methodologies and standardized baselines >>

SECTORAL SCOPE: 01, Energy industries (Renewable/Non-renewable sources)

TYPE: I–Renewable Energy Projects

CATEGORY: AMS-I.D., Grid connected renewable electricity generation, Version 18.0[7]

C.2. Applicability of methodologies and standardized baselines >>

The project activity is wind based renewable energy source, zero emission power project connected to the Gujarat state grid, which forms part of the Indian grid. The project activity will displace fossil fuel-based electricity generation that would have otherwise been provided by the operation and expansion of the fossil fuel-based power plants in Indian grid.

The approved consolidated baseline and monitoring methodology "AMS-I.D., Grid connected renewable electricity generation, Version 18" is the choice of the baseline and monitoring methodology. The applicability conditions of the methodology are discussed below:

Applicability Condition	Justification
1. This methodology is applicable to project activities that: a) Install a Greenfield plant; b) Involve a capacity addition in (an) existing plant(s) c) Involve a retrofit of (an) existing plant(s) d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or e) Involve a replacement of (an) existing plant(s).	The project activity involves installation of greenfield wind power generation plant. Hence the methodology is applicable to the project activity.
2) Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology: a) The project activity is implemented in an existing reservoir with no change in the volume of reservoir; b) The project activity is implemented in an existing reservoir, where the volume of reservoir is increased and the power density of the project activity, as per definitions given in the project emissions section, is greater than 4 W/m^2; c) The project activity results in new reservoirs and the power density of the power plant, as per definitions given in the project emissions section, is greater than 4 W/m^2.	The project activity is NOT a hydro power project. Hence the condition does not apply.
3) If the new unit has both renewable and non-renewable components (e.g. a wind/diesel unit), the eligibility limit of 15 MW for a small-scale CDM project activity applies only to the renewable component. If the new unit co-fires fossil fuel, the capacity of the entire unit shall not exceed the limit of 15 MW.	The project activity only has renewable component (i.e. wind power) of 5.5 MW and hence meets the applicability condition.

[7] AMS I.D.

4) Combined heat and power (co-generation) systems are not eligible under this category.	The project activity is a greenfield wind power generation project and hence this condition does not apply.
5) In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15 MW and should be physically distinct from the existing units.	The project activity is a greenfield project and NOT a capacity addition project. Hence the condition does not apply.
6) In the case of retrofit, rehabilitation or replacement, to qualify as a small-scale project, the total output of the retrofitted, rehabilitated or replacement power plant/unit shall not exceed the limit of 15 MW.	The project activity is a greenfield project. Hence the condition does not apply.
7) In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to a grid, then the baseline for the electricity component shall be in accordance with procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration other applicable Type-I methodologies such as "AMS-I.C.: Thermal energy production with or without electricity" shall be explored.	The project activity is a wind power project. Hence the condition does not apply.
8) In case biomass is sourced from dedicated plantations, the applicability criteria in the tool "Project emissions from cultivation of biomass" shall apply.	The project activity is Neither a fossil fuel switch project nor a biomass fired power plant. Hence the condition does not apply.

C.3 Applicability of double counting emission reductions >>

There is no double counting of emission reductions in this project activity due to the following reasons:

- The project is uniquely identifiable based on its specific location coordinates,
- Project has a dedicated commissioning certificate and a unique grid connection point,
- Project is monitored using dedicated energy meters, which are exclusively linked to the consumption point of the project developer.

C.4. Project boundary, sources and greenhouse gases (GHGs)>>

As per applicable methodology AMS-I.D, Grid connected renewable electricity generation, Version 18, "The spatial extent of the project boundary includes the project power plant and all power plants connected physically to the electricity system that the project power plant is connected to".

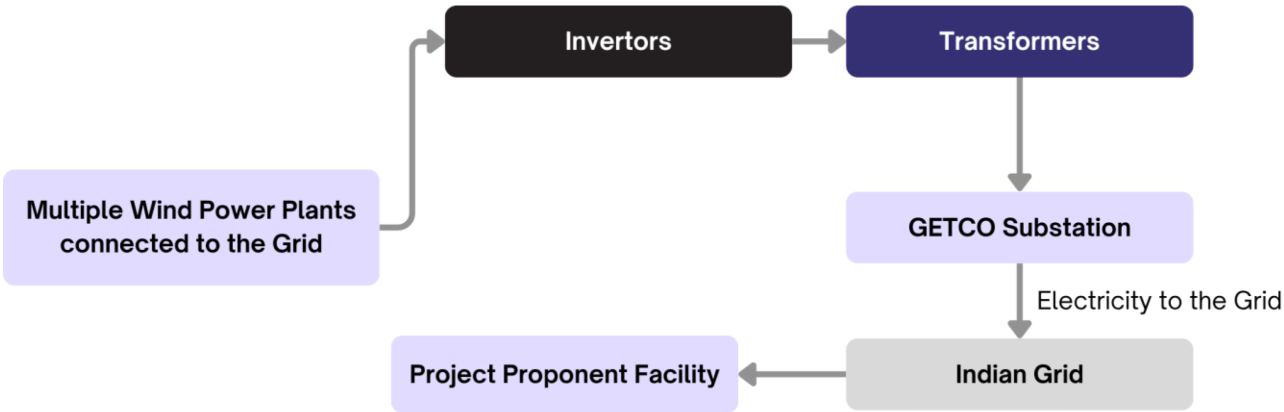

Figure 7: Project Boundary

Thus, the project boundary includes the Wind Turbine Generators (WTGs) and the Indian grid system.

Source		Gas	Included?	Justification/ Explanation
Baseline	Grid connected electricity generation	**CO_2**	**YES**	**Main emission source**
		CH_4	NO	Minor emission source
		N_2O	NO	Minor emission source
		Other	NO	No other GHG emissions were emitted from the project
Project	Greenfield Wind Power Project Activity	CO_2	NO	No CO_2 emissions are emitted from the project
		CH_4	NO	Project activity does not emit CH_4
		N_2O	NO	Project activity does not emit N_2O
		Other	NO	No other emissions are emitted from the project

C.5. Establishment and description of baseline scenario (UCR Protocol) >>

This section provides details of emission displacement rates/ coefficients/ factors established by the applicable methodology selected for the project.

As per the approved consolidated methodology AMS–I.D, Grid connected renewable electricity generation, Version 18, if the project activity is the installation of a new grid-connected renewable power plant/ unit, the baseline scenario is the following:

"The baseline scenario is that the electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid".

The project activity involves setting up of a new wind power plant to harness the green power from wind energy and to use for captive purpose via grid interface through wheeling arrangement. In the absence of the project activity, the equivalent amount of power would have been supplied by the Indian grid, which is fed mainly by fossil fuel fired plants. The power produced at grid from the other conventional sources which are predominantly fossil fuel based. Hence, the baseline for the project activity is the equivalent amount of power produced at the Indian grid.

A "grid emission factor" refers to a CO_2 emission factor (tCO_2/MWh) which will be associated with each unit of electricity provided by an electricity system. The UCR recommends an emission factor of 0.9 tCO_2/MWh for the 2013–2023 years as a fairly conservative estimate for Indian projects not previously verified under any GHG program. The UCR recommends a grid emission factor of 0.757 tCO_2/MWh for the 2024 vintage year as a fairly conservative estimate for Indian projects not previously verified under any GHG program.

In this project activity, the emission factor is determined in two distinct phases. For the period year up to 2023, a grid emission factor of 0.9 tCO_2/MWh is applied and for the year 2024, a grid emission factor of 0.757 tCO_2/MWh in accordance with the updated UCR guidelines.[8]

Net GHG Emission Reductions and Removals

Thus,

$ER_y = BE_y – PE_y – LE_y$

Where:

ER_y	=	Emission reductions in year y (tCO_2/y)
BE_y	=	Baseline Emissions in year y ($t\ CO_2/y$)
PE_y	=	Project Emissions in year y ($t\ CO_2/y$)
LE_y	=	Leakage Emissions in year y ($t\ CO_2/y$)

Baseline Emissions

Baseline emissions include only CO_2 emissions from electricity generation in power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants. The baseline emissions are to be calculated as follows:

$BE_y = EG_{PJ,y} \times EF_{grid,y}$

Where;

BE_y	=	Baseline Emissions in year y ($t\ CO_2$)
$EG_{PJ,y}$	=	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh)
$EF_{grid,y}$	=	CO_2 emission factor of grid electricity for the given year y. A factor of 0.861 tCO_2/MWh has been considered based on the latest database of the Central Electricity Authority (CEA), India. The reference for this emission factor is provided in Section C.10 (Monitoring Plan).

Baseline Emissions Table	
Year	**(tCO_2e)**
2019	5,811
2020	6,008
2021	7,059
2022	5,546
2023	3,872

[8] UCR Standard Update

2024	4,412

Project Emissions

As per paragraph 39 of AMS–I.D., Grid connected renewable electricity generation, Version 18, only emission associated with the fossil fuel combustion, emission from operation of geo-thermal power plants due to release of non-condensable gases, emission from water reservoir of Hydro should be accounted for the project emission. Since the project activity is a wind power project, project emission for renewable energy plant is nil.

Thus, $\mathbf{PE_y = 0}$.

Leakage Emissions

As per paragraph 42 of AMS–I.D, Grid connected renewable electricity generation, Version 18, 'If the energy generating equipment is transferred from another activity, leakage is to be considered'. In the project activity, there is no transfer of energy generating equipment and therefore the leakage from the project activity is considered as zero.

Hence, $\mathbf{LE_y = 0}$.

Thus, $ER_y = BE_y - PE_y - LE_y$

Total Emission reduction by the project for the current monitoring period is calculated as below:

Year	Baseline Emissions (tCO_2e)	Project Emissions (tCO_2e)	Leakage Emissions (tCO_2e)	Emission Reduction (tCO_2e)
2019	5,811	0	0	5,811
2020	6,008	0	0	6,008
2021	7,059	0	0	7,059
2022	5,546	0	0	5,546
2023	3,872	0	0	3,872
2024	4,412	0	0	4,412

B.6. Prior History>>

The project activity is a bundle of wind machines. Following are the key details under the prior history of the project:

a) The project was not applied under any other GHG mechanism. Hence project will not cause double accounting of carbon credits (i.e., COUs).

C.6. Prior History>>

The project was not applied under any other GHG mechanism. Hence project will not cause double accounting of carbon credits (i.e., COUs)

C.7. Monitoring period number and duration>>

First Monitoring Period: 5 years, 8 months, 14 days (17/04/2019 to 31/12/2024)

C.8. Changes to start date of crediting period >>

There is no change in the start date of the crediting period. The crediting period started on 17/04/2019 and remains unchanged.

C.9. Permanent changes from PCN monitoring plan, applied methodology or applied standardized baseline >>

There are no permanent changes from the registered Project Concept Note (PCN) monitoring plan and the applied methodology. The project continues to follow the originally approved monitoring approach and methodology without modifications.

C.10. Monitoring plan>>

Parameters for monitoring during the crediting period.

Data / Parameter Table 1

Data/Parameter	$EF_{grid,y}$
Data unit	tCO_2 /MWh
Description	CO2 emission factor per unit of energy of the fossil fuel used in the baseline generation source (Grid) displaced due to the project activity, during the year y
Source of data	A "grid emission factor" refers to a CO2 emission factor (tCO2/MWh) which will be associated with each unit of electricity provided by an electricity system. The UCR recommends an emission factor of 0.9 tCO_2/MWh for the 2013–2023 years as a fairly conservative estimate for Indian projects not previously verified under any GHG program. The UCR recommends a grid emission factor of 0.757 tCO2/MWh for the 2024 vintage year as a fairly conservative estimate for Indian projects not previously verified under any GHG program. UCR CoU Standard Update
Value(s) applied	Emission Factor (Till 2023): 0.90 Emission Factor (2024 Onwards): 0.757
Measurement methods and procedures	-
Monitoring frequency	-
Purpose of data	Calculation of baseline emission

Data / Parameter Table 2

<table>
<tr><td>Data / Parameter:</td><td>$\mathbf{EG_{PJ,y}}$</td></tr>
<tr><td>Data unit:</td><td>MWh</td></tr>
<tr><td>Description:</td><td>Net electricity supplied to the Indian grid facility by the project activity</td></tr>
<tr><td>Source of data:</td><td>Generation Statements</td></tr>
<tr><td>Measurement procedures (if any):</td><td>Data Type: Measured
Monitoring equipment: Energy Meters are used for monitoring
Archiving Policy: Electronic

The electricity generation is monitored directly through energy meters installed by GETCO. The generation data is recorded, maintained, and periodically revised by GETCO based on their own meter readings and calibration processes.

<table>
<tr><td>Capacity</td><td>1x1.5 MW</td><td>1x1.5 MW</td><td>1x1.25 MW</td><td>1x1.25 MW</td></tr>
<tr><td>WTG No.</td><td>M-523</td><td>M-438</td><td>B-01</td><td>B-480</td></tr>
<tr><td>Meter No.</td><td>GJU04440</td><td>XD501471</td><td>GJB01111</td><td>GJB01377</td></tr>
</table>
</td></tr>
<tr><td>Monitoring frequency:</td><td>Monthly</td></tr>
<tr><td>Value applied</td><td>37,303.69 MWh</td></tr>
<tr><td>QA/QC procedures:</td><td>Continuous monitoring, hourly measurement monthly recording. Tri-vector (TVM)/ABT energy meters with accuracy class 0.2s</td></tr>
<tr><td>Purpose of data:</td><td>Calculation of baseline emission</td></tr>
<tr><td>Any comment:</td><td>The renewable power generated by the project is wheeled for captive consumption. Therefore, during monitoring and verification, the provisions outlined in the wheeling agreement may be referred to as supporting documentation.</td></tr>
</table>